

BRAC University
Department of Computer Science and Engineering

Examination: Quiz 01
Duration: 25 Minutes

Semester : Spring 25
Full Marks: 15

CSE420: Compiler Design

Name: Solution	ID:	Section:07
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1. Define Tokens, Patterns, Lexemes (3 x 2 = 6)
2. Difference between a compiler and an interpreter. (4)
3. Find out how many lexemes are there in the below code? How many tokens will be generated for the below code? (5)

```
int a;  
a = 8;  
ab[10] = a + ab[2];
```

Question 1:

Tokens: A token is a pair consisting of a token name and an optional attribute value. The token name is an abstract symbol representing a kind of lexical unit, e.g., a particular keyword, or a sequence of input characters denoting an identifier. The token names are the input symbols that the parser processes. In what follows, we shall generally write the name of a token in boldface. We will often refer to a token by its token name.

Patterns: A pattern is a description of the form that the lexemes of a token may take. In the case of a keyword as a token, the pattern is just the sequence of characters that form the keyword. For identifiers and some other tokens, the pattern is a more complex structure that is matched by many strings.

Lexemes: A lexeme is a sequence of characters in the source program that matches the pattern for a token and is identified by the lexical analyzer as an instance of that token.

Question 2:

1. A compiler takes an entire program and a lot of time to analyze the source code, whereas the interpreter takes a single line of code and very little time to analyze it.
2. A compiled code runs faster while interpreted code runs slower.
3. A compiler displays all errors after compilation. If your code has mistakes, it will not compile. But the interpreter displays errors of each line one by one.
4. Interpretation does not replace compilation completely.
5. Compilers can contain interpreters for optimization reasons like faster performance and smaller memory footprint.

Question 3:

No of lexemes: 19

No of tokens: 19